



**SYDNEY GIRLS HIGH SCHOOL
TRIAL HIGHER SCHOOL CERTIFICATE**

2000

MATHEMATICS

**3 UNIT (Additional)
and
3/4 UNIT (Common)**

Time Allowed – 2 hours
(Plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES

NAME _____

- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used
- Each question attempted should be started on a new sheet. Write on one side of the paper only

This is a trial paper ONLY. It does not necessarily reflect the format or the contents of the 2000 HSC Examination Paper in this subject

Question 1

- (a) Find $\int_0^{0.4} \frac{3dx}{4+25x^2}$ 2
- (b) At the Sydney 2000 Olympic Games the semi-finals of the mens 100m freestyle consists of 9 swimmers wearing full body wetsuits and 7 swimmers wearing normal swimwear. How many groups of 8 swimmers, containing exactly 5 swimmers wearing full-bodied wetsuits, can be in the final? 2
- (c) If $\sin \alpha = \frac{3}{4}$ $0 < \alpha < \frac{\pi}{2}$
and $\sin \beta = \frac{2}{3}$ $\frac{\pi}{2} < \beta < \pi$
Find the exact value of:
(i) $\tan 2\alpha$
(ii) $\cos(\alpha - \beta)$ 4
- (d) Solve the equation
 $2 \ln(3x + 1) - \ln(x + 1) = \ln(7x + 4)$ 4

Question 2

- (a) Use the substitution $u = 2 - x$ to evaluate $\int_{-1}^2 x \sqrt{2-x} dx$ 4
- (b) (i) Find the value of x such that $\sin^{-1} x = \cos^{-1} x$
(ii) On the same axes sketch the graph of $y = \sin^{-1} x$ and $y = \cos^{-1} x$
(iii) On the same diagram as the graphs in (ii) draw the graph of $y = \sin^{-1} x + \cos^{-1} x$ 4
- (c) Solve $\frac{2}{3-x} \geq x$ 4

Question 3

- (a) (i) Show that the equation $\log_e x - \cos x = 0$ has a root between $x = 1$ and $x = 2$
- (ii) By taking 1.2 as the first approximation, use 1 step of Newton's method to find a better approximation to this root correct to 2 decimal places

3

- (b) Prove by mathematical induction that:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$$

4

- (c) Consider the binominal expansion of $(3 + 2x)^{11}$

- (i) Let T_k be the k th term in the expansion (where the terms are written out in increasing powers of x) Show that

$$\frac{T_{k+1}}{T_k} = \frac{2x(12-k)}{3k}$$

- (ii) Find the greatest coefficient in the expansion.

5

Question 4

- (a) A spherical metal ball is being heated such that the volume increases at a rate of $2\pi \text{ mm}^3/\text{min}$. At what rate is the surface area increasing when the radius is 3mm? 3
- (b) A is the point $(-4,1)$ and B is the point $(2,4)$. Q is the point which divides AB internally in the ratio 2:1 and R is the point which divides AB externally in the ratio 2:1. P (x,y) is a variable point which moves so that $PA = 2PB$.
- (i) find the co-ordinates of Q and R
- (ii) show that the locus of P is a circle on QR as diameter. 5
- (c) At any time t the rate of cooling of the temperature T of a body when the surrounding temperature is P , is given by the equation.

$$\frac{dT}{dt} = -k(T-P) \text{ for some constant } k$$

- (i) Show that the solution $T = P + Ae^{-kt}$ for some constant A satisfies this equation
- (ii) A metal bar has a temperature of 1340° and cools to 1010° in 12 minutes when the surrounding temperature is 25°C . Find how much longer it will take the bar to cool to 60°C , giving your answer correct to the nearest minute 4

Question 5

- (a) (i) Prove $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ where v denotes velocity 6
- (ii) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$ where x is the displacement from O. The initial velocity of the particle is 2m/s at O
- a) Show that $v^2 = 4e^{-x}$
- b) Describe the subsequent motion of the particle making reference to its speed and direction.
- (b) Consider the binominal expansion 3
- $$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$
- (i) Use a suitable substitution to find the value of $\binom{n}{0} + 2\binom{n}{1} + 2^2\binom{n}{2} + \dots + 2^n\binom{n}{n}$
- (ii) Differentiate both sides of the identity and then use a suitable substitution to find the value of $\binom{n}{1} - 2\binom{n}{2} + 3\binom{n}{3} - \dots + (-1)^{n-1}n\binom{n}{n}$
- (c) Write $2 \cos \theta + \sin \theta$ in the form $A \cos(\theta - \alpha)$. Hence solve $2 \cos \theta + \sin \theta = \sqrt{5}$ $0 \leq \theta \leq 2\pi$: 3

Question 6

- (a) (i) Using long division divide the polynomial $f(x) = x^4 - x^3 + x^2 - x + 1$ by the polynomial $d(x) = x^2 + 4$.
Express your answer in the form $f(x) = d(x).q(x) + r(x)$

- (ii) Hence find the values of the constants a and b so that $x^4 - x^3 + x^2 + ax + b$ is
Divisible by $x^2 + 4$

3

- (b) Find the volume of revolution formed when the area bounded by the x axis
and the curve $y = \cos x$ between $x = \frac{-\pi}{2}$ and $x = \frac{\pi}{2}$ is rotated about the x axis

4

- (c) A competitor shoots an arrow with velocity 20m/s^{-1} to hit a target at a horizontal distance
20m from the point of projection and a height of 10m above the ground

- (i) Using calculus prove that the co-ordinates of the arrow at time t are
given by

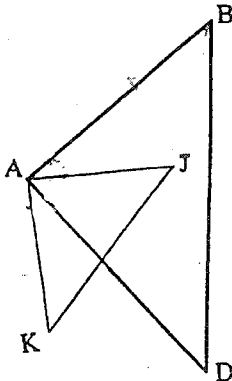
$$\begin{aligned}x &= 20t \cos \alpha \\y &= -5t^2 + 20t \sin \alpha\end{aligned}$$

- (ii) Find two possible angles of projection ($g = 10\text{m/s}$)

5

Question 7.

- (a) ABD and AJK are two isosceles triangles both right angled at A



Copy the diagram onto your answer sheet

6

- (i) Show that $\hat{BJA} = \hat{DKA}$
- (ii) BJ is produced to meet DK at X. Show that $BX \perp DK$
- (ii) The square ABCD is completed. Show that $\hat{BXC} = 45^\circ$

- (b) A ship needs 7.5m of water to pass down a channel safely. At high tide the channel is 9m deep and at low tide the channel is 3m deep.
High tide is at 4:00am
Low tide is at 10:20 am.
Assume that the tide rises and falls in Simple Harmonic Motion

- (i) What is the latest time before noon, to the nearest minute, that the ship can safely proceed through the channel?
- (ii) In the 12 hours starting from 9:00 am between what times will the ship be able to proceed safely down the channel?

6

END OF PAPER

$$\begin{aligned}
 Q1a) \int_0^{0.4} \frac{3dx}{4+25x^2} &= \frac{3}{2.5} \left[\tan^{-1}\left(\frac{5x}{2}\right) \right]_0^{2/5} \\
 &= \frac{3}{10} (\tan^{-1}(1) - \tan^{-1}(0)) \\
 &= \frac{3}{10} \cdot \frac{\pi}{4} \\
 &= \frac{3\pi}{40}
 \end{aligned}$$

$$\begin{array}{ccc}
 9\text{L6}, & 7\text{H} & \Rightarrow 9\text{C}_5 \times 7\text{C}_3 = 4410 \\
 \downarrow & \downarrow & \\
 5 & 3 &
 \end{array}$$

$$\begin{aligned}
 c) \sin \alpha &= \frac{3}{4} \quad 0 < \alpha < \pi/2 & \begin{array}{c} \text{4} \\ \text{3} \\ \text{5} \end{array} \\
 \sin \beta &= \frac{4}{5} \quad \pi/2 < \beta < \pi & \begin{array}{c} \text{3} \\ \text{2} \\ \text{5} \end{array}
 \end{aligned}$$

$$\begin{aligned}
 i) \tan 2\alpha &= \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{2 \times \frac{3}{4}}{1 - \frac{9}{16}} \times \frac{1}{1 - \frac{9}{16}} \\
 &= \frac{6}{4} \times \frac{16}{7} \\
 &= -2\sqrt{7}
 \end{aligned}$$

$$\begin{aligned}
 ii) \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\
 &= \frac{4}{5} \cdot \frac{3}{5} + \frac{3}{4} \cdot \frac{4}{5} \\
 &= \frac{12}{25} + \frac{12}{25} = \frac{24}{25}
 \end{aligned}$$

$$d) 2 \ln(3x+1) - \ln(x+1) = \ln(7x+4)$$

$$\therefore (3x+1)^2 = (7x+4)(x+1)$$

$$9x^2 + 6x + 1 = 7x^2 + 11x + 4$$

$$\therefore 2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$\therefore x = -\frac{1}{2}, 3$$

$$\text{at } x = -\frac{1}{2}, \ln(3x+1) \text{ is undefined} \\
 \therefore x = 3$$

Q 2 a) $\int_{-1}^2 x \sqrt{2-x} dx$ $u = 2-x$ $x = -1, u = 3$
 $du = -dx$ $x = 2, u = 0$
 $x = 2-u$

$$= \int_3^0 (2-u) \sqrt{u} (-du)$$

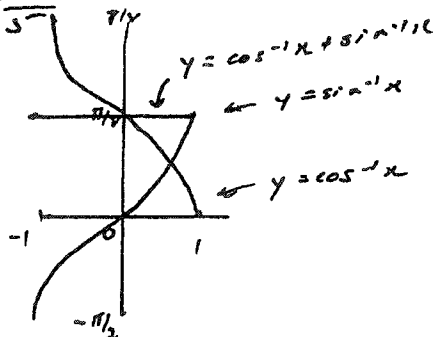
$$= \int_0^3 (2u^{1/2} - u^{3/2}) du$$

$$= \left[\frac{4}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^3$$

$$= \frac{4}{3} \cdot 3\sqrt{3} - \frac{2}{5} \cdot 9\sqrt{3}$$

$$= \frac{2\sqrt{3}}{5}$$

b) i)



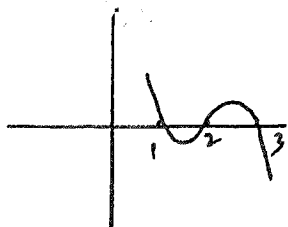
At $\sin^{-1} x = \cos^{-1} x$

$\therefore x = 1/2$

c) $\frac{2}{3-x} > x$
 $\therefore \frac{2(3-x)^2}{3-x} > x(3-x)^2$, $x \neq 3$

$\therefore 2(3-x) > x(3-x)^2$
 $\therefore x(3-x)^2 - 2(3-x) \leq 0$
 $(3-x) \{ x(3-x) - 2 \} \leq 0$
 $(3-x) (3x-x^2-2) \leq 0$
 $(3-x) (x^2-3x+2) \neq 0$
 $(3-x) (x-1)(x-2) \neq 0$, $x \neq 3$

$\therefore x \leq 1, 2 \leq x < 3$



3 a) let $f(x) = \ln x - \cos x$

$$f(1) = \ln 1 - \cos 1 = -0.54 < 0$$

$$f(2) = \ln 2 - \cos 2 = 1.11 > 0$$

$\therefore$ since $f(x)$ changes sign,
there is a root $1 < x < 2$.

ii $f'(x) = \frac{1}{x} + \sin x$

$$f(1.2) = -0.18$$

$$f'(1.2) = 1.765$$

$$x_0 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.2 + \frac{0.18}{1.765}$$

$$= 1.30$$

b) Prove $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$

at $n=1$, LHS = $\frac{1}{1 \times 2} = \frac{1}{2}$, RHS = $1 - \frac{1}{1+1} = \frac{1}{2} = \text{LHS}$

$\therefore$ true for $n=1$

assume true for $n=k$, i.e. assume $\frac{1}{1 \times 2} + \dots + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1}$

& prove for $n=k+1$, i.e. prove $\frac{1}{1 \times 2} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{k+2}$

Now LHS = $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$

$$= 1 - \frac{1}{k+1} + \frac{1}{(k+1)(k+2)} \quad \text{using assumption}$$

$$= 1 - \frac{k+2-1}{(k+1)(k+2)}$$

$$= 1 - \frac{1}{k+2} = \text{RHS}$$

$\therefore$ if true for $n=k$ it is true for $n=k+1$.

since true for $n=1$ it is thus true for $n=2$, then $n=3$ & so on for all positive integral n .

c) $(3+2x)^n$ $T_{k+1} = {}^nC_k a^{n-k} b^k = {}^nC_k 3^{n-k} (2x)^k$

$$T_k = {}^nC_{k-1} a^{n-(k-1)} b^{k-1} = {}^nC_{k-1} 3^{n-(k-1)} (2x)^{k-1}$$

$$\therefore \frac{T_{k+1}}{T_k} = \frac{11!}{(11-k)!k!} \cdot \frac{3^{n-k} (2x)^k}{11!} \cdot \frac{(12-k)!(k-1)!}{11!} \cdot \frac{1}{3^{12-k} (2x)^{k-1}}$$

$$= \frac{12-k}{k} \cdot \frac{2x}{3}$$

For greatest, coefft $\frac{T_{k+1}}{T_k} > 1$, $\therefore \begin{aligned} 2(12-k) &> 3k \\ 24-2k &> 3k \\ 5k &< 24 \\ \therefore k &= 4 \end{aligned}$

Question 4

a) Given $\frac{dV}{dt} = 2\pi \text{ mm}^3/\text{min}$ find $\frac{dA}{dt}$ when $r = 3$

$$V = \frac{4}{3}\pi r^3, \quad A = 4\pi r^2$$

$$\frac{dV}{dr} = 4\pi r^2, \quad \frac{dA}{dr} = 8\pi r$$

3

$$\frac{dA}{dt} = \frac{dV}{dt} \times \frac{dr}{dV} \times \frac{dA}{dr}$$

$$= 2\pi r \times \frac{1}{4\pi r^2} \times 8\pi r$$

$$= \frac{4\pi}{r} \quad \text{when } r = 3$$

$$\frac{dA}{dt} = \frac{4\pi}{3} \text{ mm}^2/\text{min}$$

b) i) Coords of C $x = \frac{2(2) + 1(-4)}{3}, y = \frac{2(4) + 1(1)}{3}$

$$x = 0$$

$$y = 3$$

2

Coords of R $x = \frac{2(2) - 1(-4)}{2-1}, y = \frac{2(4) - 1(1)}{2-1}$

$$x = 8$$

$$y = 7$$

ii) $PA = 2PB$

$$\sqrt{(x+1)^2 + (y-1)^2} = 2\sqrt{(x-2)^2 + (y-4)^2}$$

$$x^2 + 8x + 16 + y^2 - 2y + 1 = 4[x^2 - 4x + 4 + y^2 - 8y + 16]$$

2

$$3x^2 + 3y^2 - 24x - 30y + 63 = 0$$

$$x^2 - 8x + y^2 - 10y = -21$$

$$(x-4)^2 + (y-5)^2 = -21 + 16 + 25$$

$$(x-4)^2 + (y-5)^2 = 20 \quad \text{is circle centre } (4, 5), r = \sqrt{20}$$

1

Midpt of R $x = \frac{0+8}{2}, y = \frac{3+7}{2}$

$$= 4$$

$$= 5$$

is centre circle

Radius of R $r = \sqrt{(4-0)^2 + (5-3)^2}$

$$= \sqrt{20}$$

1

c) i) $T = P + Ae^{-kt} \Rightarrow T - P = Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T - P)$$

ii) initially $t = 0, T = 1340, P = 25$

$$1340 = 25 + A \quad A = 1315$$

$$T = 25 + 1315e^{-kt}$$

when $t = 12, T = 1010$

$$1010 = 25 + 1315e^{-12k}$$

$$e^{-12k} = \frac{197}{1315}$$

3

$$k = \left[\log_e \left(\frac{197}{1315} \right) \right] \div -12 \quad (k = 0.024 \dots)$$

when $T = 60$

$$60 = 25 + 1315e^{-kt}$$

$$t = \log_e \left(\frac{1}{263} \right) \div k$$

Question Five

1) R.T.P $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

Now $a = \frac{dv}{dt} \cdot \frac{dx}{dt}$ (chain rule)
 $= v \frac{dv}{dx}$ ($v = \frac{dx}{dt}$)
 $= \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$
 $= \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

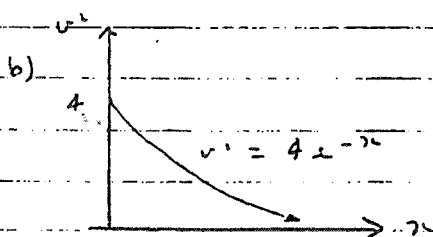
ii) a) $\ddot{x} = -2e^{-x}$

$\therefore \frac{1}{2} v^2 = \int -2e^{-x} dx$

$\frac{1}{2} v^2 = 2e^{-x} + C$

Initially $x=0, v=2, C=0$

$\therefore v^2 = 4e^{-x}$



velocity in positive direction and decreasing

b) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$

i) put $x=2$

$3^n = \binom{n}{0} + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^n\binom{n}{n}$

ii) d.w.r.t. x

$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + \dots + n\binom{n}{n}x^{n-1}$

put $x=-1$

$0 = \binom{n}{1} - 2\binom{n}{2} + \dots + (-1)^{n-1}n\binom{n}{n}$

c) $2\cos\theta + \sin\theta = A\cos(\theta-\alpha) = \sqrt{5}$

$2\cos\theta + \sin\theta = A[\cos\theta\cos\alpha - \sin\theta\sin\alpha] = \sqrt{5}$

divide by $A = \sqrt{5}$

$\frac{2}{\sqrt{5}}\cos\theta + \frac{1}{\sqrt{5}}\sin\theta = \cos\theta\cos\alpha - \sin\theta\sin\alpha = 1$

ie $\cos\alpha = \frac{2}{\sqrt{5}}$
 $\sin\alpha = \frac{1}{\sqrt{5}}$ } α acute

$\tan\alpha = \frac{1}{2}, \alpha = 0.46\dots$

Now $\cos(\theta-\alpha) = 1 \therefore \theta-\alpha = 0, 2\pi$

∴ i)

$$\begin{array}{r}
 x^2 - x - 3 \\
 x^2 + 4 \overline{) x^4 - x^3 + x^2 - x + 1} \\
 \underline{x^4 + 4x^2} \\
 -x^3 - 3x^2 - x \\
 \underline{-x^3 - 4x} \\
 -3x^2 + 3x + 1 \\
 \underline{-3x^2 - 12} \\
 3x + 13
 \end{array}$$

or

$$\begin{array}{r}
 x^2 - x - 3 \\
 x^2 + 4 \overline{) x^4 - x^3 + x^2 + ax + b} \\
 \underline{x^4 + 4x^2} \\
 -x^3 - 3x^2 + ax \\
 \underline{-x^3 - 4x} \\
 -3x^2 + x(a+4) + b \\
 \underline{-3x^2 - 12} \\
 x(a+4) + b + 12
 \end{array}$$

Remainder = 0

$$\begin{aligned}
 \therefore a+4 &= 0 & a &= -4 \\
 b+12 &= 0 & b &= -12
 \end{aligned}$$

$$f(x) = (x^2 + 4)(x^2 - x - 3) + 3x + 13$$

∴,

$$f(x) - 3x - 13 = (x^2 + 4)(x^2 - x - 3)$$

$$x^4 - x^3 + x^2 - x + 1 - 3x - 13$$

$$= x^4 - x^3 - x^2 - 4x - 12$$

$$\therefore a = -4, b = -12$$



$$\begin{aligned}
 y &= \cos x \\
 y &= \cos^2 x \\
 y &= \frac{1}{2}(\cos 2x + 1)
 \end{aligned}$$

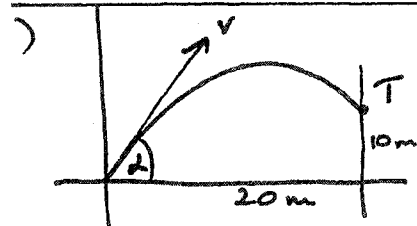
$$V = \pi \int y^2 dx$$

$$= \pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}(\cos 2x + 1) dx$$

$$= \frac{\pi}{2} \times 2 \int_0^{\frac{\pi}{2}} \cos 2x + 1 dx$$

$$= \pi \left[\frac{\sin 2x}{2} + x \right]_0^{\frac{\pi}{2}} = \pi \left[0 + \frac{\pi}{2} - 0 \right]$$

$$\therefore Vol = \frac{\pi^2}{2} u^3$$



$$t=0, x=0, y=0$$

$$v=20$$

$$\dot{x} = 20 \cos \alpha$$

$$\dot{y} = 20 \sin \alpha$$

ii) when $x=20, y=10$

$$\text{i.e. } 20 = 20t \cos \alpha \Rightarrow t = \frac{1}{\cos \alpha}$$

$$\text{and } 10 = -5t^2 + 20t \sin \alpha$$

$$i) \ddot{x} = 0$$

$$\dot{x} = c_1 \quad (t=0, \dot{x}=20 \cos \alpha)$$

$$\dot{x} = 20 \cos \alpha$$

$$x = 20t \cos \alpha + c_2$$

$$x=0 \quad t=0 \therefore c_2=0$$

$$\therefore x = 20t \cos \alpha$$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c_3$$

$$20 \sin \alpha = c_3$$

$$\therefore \dot{y} = -gt + 20 \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + 20t \sin \alpha$$

$$c_4 = 0 \quad (\text{when } t=0, y=0)$$

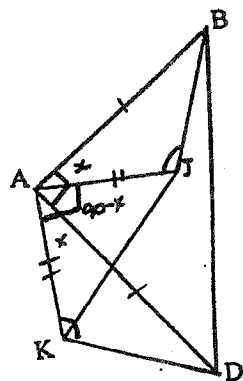
$$\therefore y = -\frac{1}{2}gt^2 + 20t \sin \alpha$$

$$= -5t^2 + 20t \sin \alpha$$

$$\tan^2 \alpha - 4 \tan \alpha + 3 = 0$$

$$(\tan \alpha - 3)(\tan \alpha - 1) = 0$$

2)



$$AB = AD$$

$$\text{let } \angle BAJ = x$$

$$AJ = AK$$

$$\therefore \angle JAD = 90 - x \quad (\text{adj. compl. } \angle s)$$

$$\angle BAD = 90^\circ$$

$$\text{also } \angle KAD = x \quad (\text{adj. comp. } \angle s)$$

$$\angle JAK = 90^\circ$$

Now in Δs BAJ and DAK

$$AB = AD \quad (\text{equal sides of isosce } \Delta BAD)$$

$$AJ = AK \quad (\text{ " " " isosce } \Delta AJK)$$

$$\angle BAJ = \angle KAD \quad (\text{proven above})$$

$$\therefore \Delta BAJ \cong \Delta DAK \quad (SAS)$$

$$\text{Hence, } \angle BJA = \angle DKA \quad (\text{corresp } \angle s \text{ of congr. } \Delta s)$$

$$\text{ii) } \angle AJB + \angle AJX = 180^\circ \quad (\text{adj. suppl. } \angle s)$$

$$\therefore \angle AJX = 180 - \angle BJA$$

$$\text{Now, } \angle JAK + \angle AKX + \angle KXJ + \angle AJX = 360^\circ \quad (\angle \text{ sum of quad})$$

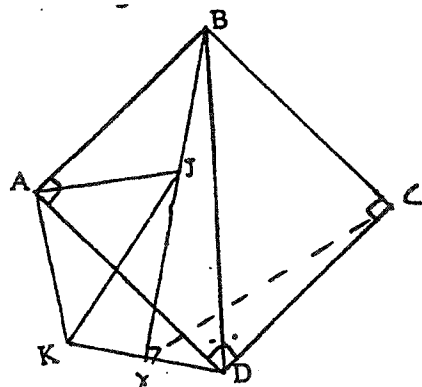
$$90^\circ + \angle AKX + \angle KXJ + 180 - \angle BJA = 360$$

$$\angle KX = \angle DKA \quad \text{ie } \angle BJA + \angle KXJ - \angle BJA = 90^\circ$$

$$= \angle BJA$$

$$\therefore \angle KXJ = 90^\circ \quad \therefore JX \perp KD$$

iii)



$$\text{since } \angle BCD = 90^\circ \text{ and } \angle BXD = 90^\circ$$

then BCDX are concyclic with BD a diameter.

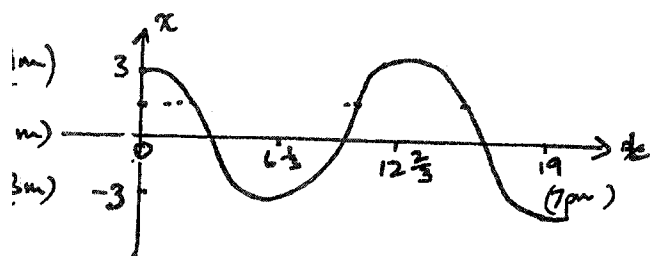
Now, BD bisects $\angle ADC$ (diagonal of square)

$$\therefore \angle BDC = 45^\circ$$

$$\text{and } \angle BDC = \angle BXC \quad (\angle s \text{ on same arc,})$$

$$\therefore \angle BXC = 45^\circ$$

7(b) high tide = 9m at 4am
low tide = 3m at 10.20am



$$x = 3 \cos \pi t$$

$$= 3 \cos \frac{3\pi t}{19}$$

$x = 1.5$ (ie 7.5m deep)

$$1.5 = 3 \cos \frac{3\pi t}{19}$$

$$\frac{1}{2} = \cos \frac{3\pi t}{19}$$

$$\therefore \frac{3\pi t}{19} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \dots$$

$$t = \frac{19}{9}, \frac{95}{9}, \frac{133}{9}, \dots$$

ie after $\frac{19}{9}$ hours

$$\Rightarrow 4\text{am} + 2\text{h } 6\text{min}$$

$$6.06\text{am}$$

$x \geq 1.5$ (from graph)

$$0 \leq t \leq \frac{19}{9}, \quad \frac{95}{9} \leq t \leq \frac{133}{9}$$

ie between 4am and 6.06am

and 4am + 10h 33min and 4am + 14h 46min

between 2.33pm to 6.46pm

let 6m be equilibrium (ie $x=0$)

$\therefore$ high tide $x=3$

low tide $x=-3$

let $t=0$ be at 4am

$\therefore t=6\frac{1}{2}$ is at 10.20am

$\therefore$ period = $12\frac{2}{3} \Rightarrow \pi = \frac{3}{19}$

amplitude = 3